1. Notation

We work with

$$G := \mathrm{PGL}_2(\mathbb{R}), \quad \Gamma := \mathrm{PGL}_2(\mathbb{Z}),$$
$$n(x) := \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \in N := \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \leq G,$$
$$a(y) := \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix}.$$

2. Fourier expansions of automorphic functions

A nice enough function $f : \mathbb{R}/\mathbb{Z} \to \mathbb{C}$ has a Fourier expansion $f(x) = \sum_n e(nx)\hat{f}(n)$, with Fourier coefficients $\hat{f}(n) = \int_{\mathbb{R}/\mathbb{Z}} f(x)e(-nx)\,dx$.

Given an automorphic function

$$\varphi : \mathrm{PGL}_2(\mathbb{Z})\backslash \mathrm{PGL}_2(\mathbb{R}) \to \mathbb{C},$$

we may view the function $\varphi(\bullet g) : N(\mathbb{Z})\backslash N(\mathbb{R}) \to \mathbb{C}$ as an a function on $\mathbb{R}/\mathbb{Z}$, so it admits a Fourier expansion:

$$\varphi(n(x)g) = \sum_{n\in\mathbb{Z}} e(nx)W_{\varphi,n}(x),$$

where

$$W_{\varphi,n}(x) := \int_{\mathbb{R}/\mathbb{Z}} \varphi(n(u)g)e(-nu)\,du.$$

If φ is a Hecke eigenform, then we can write

$$W_{\varphi,n}(g) = \frac{\lambda(n)}{|n|^{1/2}} W_{\varphi,1}(a(n)g), \tag{1}$$

where $\lambda(n)$ is the Hecke eigenvalue. Thus

$$\varphi(g) = \sum_{n\in\mathbb{Z}_{\neq 0}} \frac{\lambda(n)}{|n|^{1/2}} W_{\varphi}(a(n)g). \tag{2}$$

Example 1. Classical holomorphic modular form f of weight k.

$$f(x+iy) = \sum_{n\geq 1} a_n e(nz), \quad e(z) := e^{2\pi i z}.$$

$$y^{k/2}f(x+iy) = \sum_{n\geq 1} a_n n^{-k/2} W_k(ny)e(nx), \quad W_k(y) := y^{k/2}e^{-2\pi y},$$

$$a_n n^{-k/2} = \frac{\lambda(n)}{|n|^{1/2}}.$$

If we define $\varphi : \mathrm{PGL}_2(\mathbb{Z})\backslash \mathrm{PGL}_2(\mathbb{R}) \to \mathbb{C}$ by $\varphi(g) := f|_k g(i)$, where in general

$$f|_k g(z) := \frac{(\det g)^{k/2}}{(cz+d)^k} f\left(\frac{az+b}{cz+d}\right), \qquad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then, since $n(x)a(y)\cdot i = x+iy$, we have

$$y^{k/2}f(x+iy) = \varphi(n(x)a(y)),$$
$$W_k(y) = W_{\varphi,1}(a(y)).$$

Remark 2. The idea of the proof of (1) is that by "uniqueness of Whittaker functionals", we know that the two quantities $W_{\varphi,n}$ and $W_{\varphi,1}(a(n)\bullet)$ are proportional. On the other hand, by studying how Hecke operators affect Fourier (or Whittaker) expansions, we see that the former is $\frac{\lambda(n)}{|n|^{1/2}}$ times the latter. If the latter were to vanish identically, then it would follow that the former vanishes identically, and so φ would vanish. Therefore the latter does not vanish identically, and we get the desired relation. For the case of (2) relevant for Example 1, see [3, Chapter VII, §5.4, Thm 7].

3. Modularity

We restrict now to the case that φ lies inside some irreducible subrepresentation π of the space of cusp forms on $\Gamma\backslash G$, consisting of Hecke eigenfunctions with eigenvalues $\lambda(n)$. Thus $\lambda(n)$ depends only upon π, not the choice of $\varphi \in \pi$.

The theory of the Kirillov model says that for any function h on $\mathbb{R}^\times$, there is a unique $\varphi \in \pi$ so that

$$W_\varphi(a(\bullet)) := [y \mapsto W_\varphi(a(y))] = h.$$

This is useful, because we then can write

$$\varphi(1) = \sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} W_\varphi(a(n)).$$

On the other hand, φ is invariant under Γ, so for any $\gamma \in \Gamma$ (e.g., $\gamma = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$), we may write $\varphi(1) = \varphi(\gamma)$, hence

$$\sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} W_\varphi(a(n)) = \sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} W_\varphi(a(n)\gamma).$$

Remark 3. Assuming π is tempered, any Whittaker function enjoys the decay estimates

$$W_\varphi(a(y)) \ll_\varphi \min(y^{1/2+\varepsilon}, e^{-2\pi y}).$$

Then the problem arises: given a function $h : \mathbb{R}^\times \to \mathbb{C}$, having chosen φ so that $W_\varphi(a(\bullet)) = h$, can we understand the new function

$$h_\gamma := W_\varphi(a(\bullet)\gamma)?$$

This is typically understood using the local functional equation and/or Bessel transform.

A very convenient case is when we start with some fixed φ, and just use that for all $g \in G$ and $\gamma \in \Gamma$, we have $\varphi(\gamma g) = \varphi(g)$, hence

$$\sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} W_\varphi(a(n)g) = \sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} W_\varphi(a(n)\gamma g). \tag{3}$$

We can just pretend at first approximation that $W_\varphi(a(\bullet)k)$ looks like some fixed bump function for all k in some fixed compact subset of G. It's useful to recall that any element of G can be written using the Cartan decomposition as $k_1 a(y) k_2$, or using the Iwasawa decomposition as $n(x)a(y)k$.

4. Case studies

Here we give some examples of how to apply (3). We'll focus on cases where φ is *fixed*. This case is the relevant one if one would like to try (as we did in our internal seminar) understanding papers like [1] from a geometric perspective; that's the motivation for this whole discussion.

4.1. **Varying archimedean frequency.** Let's say we're given $h \in C_c^\infty(\mathbb{R}^\times)$ and we want to understand what the following sums look like:

$$\sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} h(n/Y) e(\xi n). \tag{4}$$

To do so, we choose φ so that $W_\varphi(a(\bullet)) = h$. Then we observe that the above sum is exactly

$$\varphi(n(\xi)a(1/Y)).$$

Note that the image of the argument of φ in $\mathbb{H}$ is $\xi + i/Y$.

One simple thing one can do is take $\xi = 0$ and use this to see that we get massive decay when $Y \to \infty$. How to see that? Well, the point in question, i/Y, is tending to zero. So we should apply some element $\gamma \in \Gamma$ that maps 0 to ∞. We can take

$$\gamma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then we get

$$\varphi(a(1/Y)) = \varphi(\gamma a(1/Y)) = \varphi(a(Y)\gamma).$$

By passing to Fourier expansions, we deduce that

$$\sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} h(n/Y) = \sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} W_\varphi(a(nY)\gamma).$$

Since γ lies in some fixed compact collection, we can think of $W_\varphi(a(\bullet)\gamma)$ as roughly a fixed bump function (more precisely, it decays like $y^{1/2}$ near zero and exponentially near ∞, but whatever). In particular, the above sum is essentially restricted to $nY \ll 1$. But if $Y \to \infty$ and n is a nonzero integer, then this last condition is never satisfied, and so the right hand side is vanishingly small.

4.2. **Varying non-archimedean frequency.** Another basic interesting example: for a natural number q, suppose we want to understand (4) in the special case that $\xi = a/q$, where a is an integer coprime to q. That is to say, we care about

$$\sum_{n\neq 0} \frac{\lambda(n)}{|n|^{1/2}} h(n/Y) e_q(an),$$

where $e_q : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}^\times$, $e_q(x) := e(x/q)$. Welp, with the same choice of φ as above, the above sum is

$$\varphi(n(a/q)a(1/Y)).$$

The image in $\mathbb{H}$ is $a/q + i/Y$.

Suppose we're in the regime where Y is quite large (we can figure out later exactly how large we need it to be). Then the argument of φ is tending to the cusp a/q. We should choose γ that maps this cusp to ∞. We can take

$$\gamma = \begin{pmatrix} -\bar{a} & * \\ q & -a \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Then, we try using that

$$\varphi(n(a/q)a(1/Y)) = \varphi(\gamma n(a/q)a(1/Y)).$$

This begs the question of understanding

$$\gamma n(a/q)a(1/Y).$$

Welp, if we have

$$\gamma\left(\frac{a}{q}+\frac{i}{Y}\right) = x' + iy',$$

then we know that

$$\gamma n(a/q)a(1/Y) = n(x')a(y')k$$

for some $k \in \mathrm{SO}(2)$.

We might try factoring

$$\gamma = n(-\bar{a}/q)\begin{pmatrix} 0 & -1/q \\ q & 0\end{pmatrix} n(-a/q),$$

then

$$\gamma n(a/q)a(1/Y) = n(-\bar{a}/q)\begin{pmatrix} 0 & -1/q \\ q/Y & 0\end{pmatrix} = n(-\bar{a}/q)a(Y/q^2)w,$$

where

$$w := \begin{pmatrix} 0 & -1 \\ 1 & 0\end{pmatrix}.$$

This tells us that

$$\sum_{n\neq 0}\frac{\lambda(n)}{|n|^{1/2}}h(n/Y)e_q(an) = \sum_{n\neq 0}\frac{\lambda(n)}{|n|^{1/2}}e_q(-\bar{a}n)W_\varphi(a(nY/q^2)w).$$

The right hand side is essentially restricted to $n/(q^2/Y) \ll 1$. So, for example:

- If $Y \ggg q^2$, then the dual sum (the RHS of the above) is of length $q^2/Y \lll 1$, so it is negligible.
- If $Y \approx q$, then the original sum and the dual sum are both of length roughly q, so we "gain nothing" (at least as far as trivial bounds are concerned) by dualizing (i.e., applying the above formula).

5. Varying archimedean frequency, but using Dirichlet approximation

A third example along such lines, and maybe the most relevant: take a general ξ as in (4), but let's do Dirichlet approximation on it first to approximate it by some rational a/q. We should get, for some $q \leq Q$, that

$$|\xi - a/q| \leq \frac{1}{qQ}.$$

Write $\xi = a/q + \delta$, where $|\delta| \leq 1/(qQ)$. The sum of interest may then be written

$$\varphi(n(a/q)n(\delta)a(1/Y)) = \varphi(n(a/q)a(1/Y)) = (n(Y\delta)\varphi)(n(a/q)a(1/Y)).$$

(Here we are using the notation $(g\varphi)(h) := \varphi(hg)$.)

Now we apply the above analysis, but with φ replaced by its right translate $n(Y\delta)\varphi$. This gives that the sum of interest is

$$\sum_{n\neq 0}\frac{\lambda(n)}{|n|^{1/2}}e_q(-\bar{a}n)W_\varphi(a(nY/q^2)wn(Y\delta)).$$

We are led to the problem of understanding the above argument. If it happens that $Y\delta \ll 1$, then there is no change from before. We know that $\delta \ll 1/qQ$, so this happens if $Y \ll qQ$.

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$
$$wn(-1/x) = n(x)a(x^2)wn(x)w.$$

This is useful in the limit as $x \to 0$.

$$W_\varphi(a(y)w) = \int_{t\in\mathbb{R}^\times} W_\varphi(a(t))J(ty)\, d^\times t.$$

References

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[2] E. Kowalski, P. Michel, and J. VanderKam. Rankin-Selberg L-functions in the level aspect. *Duke Math. J.*, 114(1):123–191, 2002.

[3] J.-P. Serre. *A course in arithmetic*. Springer-Verlag, New York, 1973. Translated from the French, Graduate Texts in Mathematics, No. 7.